

SUPERNOVA SMARTHOME

Complete Feature Catalogue

8

Relay Channels

120+

UI Themes

4

Control Methods

2

Hardware Modes

One ESP8266 board. Full home automation — web, voice, chat, and touch.

Ways to Control Your Home

Every switch can be controlled interchangeably from any of these — all stay perfectly in sync.

■ Web Dashboard

Mobile-friendly, login-protected control panel with instant real-time updates pushed to every open browser via WebSocket.

■ Telegram Bot

Full inline-keyboard switch panel plus text commands, with per-user access control and admin security alerts.

■ Amazon Alexa

Each switch appears as its own voice-controllable device via Philips Hue emulation — no extra Alexa skill needed.

■ Browser Voice Control

Hands-free control straight from Chrome using the Web Speech API — say "turn on all" or a switch name, no app required.

■ Physical Switches

Wire in real toggle switches for wall-mounted manual control, always available even if WiFi is down.

■ Touch Switches

TTP223 touch-sensor support for a modern, buttonless wall panel feel.

Scheduling & Automation

■ Duration Timers

Turn any switch ON or OFF automatically after a set number of minutes from now — perfect for "off in 30 minutes" style control.

■ Date & Time Timers

Schedule a switch to fire at an exact future date and time — e.g. 6:30 AM every day, or a specific one-off event.

■ Power-Cut Safe

All scheduled timers are stored in EEPROM and survive a power outage or reboot — they simply resume once power returns.

■ Smart Conflict Handling

Setting a new timer for a switch automatically cancels any conflicting timer on that same switch, so schedules never fight each other.

■ Power-On Default Modes

Choose per switch: restore Last State, always turn ON, or always turn OFF whenever the device boots or power is restored.

■ Manual Override

Toggling a switch by hand (web, Telegram, Alexa, or a physical switch) instantly cancels any timer running on it.

Hardware Flexibility

■ 5-Channel GPIO Mode

Direct GPIO relay control for compact 5-relay setups — no extra chips required.

■ 8-Channel I2C Mode

Expand to 8 relays using a single PCF8574 I/O expander over I2C, freeing up GPIO pins.

■ Software-Switchable

Change between 5-channel and 8-channel modes from Settings — no re-flashing required.

■ Automatic Pin Conflict Handling

TX/RX and onboard-LED pin sharing with switch inputs is auto-detected and safely managed by the firmware.

Networking & Updates

■ AP + Station WiFi Modes

Starts as its own Access Point for easy first-time setup, then joins your home WiFi network as a Station.

■ Static IP Support

Optionally assign a fixed IP, gateway, and subnet instead of relying on DHCP.

■ Over-the-Air (OTA) Updates

Push new firmware wirelessly from Arduino IDE — no USB cable needed once the device is on your network.

■ Session-Based Login

Secure cookie-based web authentication protects the dashboard and settings from unauthorized access.

■ QR WiFi Sharing

Generates a scannable QR code for the device's WiFi credentials — guests connect in one tap, no typing passwords.

■ Unauthorized-Access Alerts

Any unrecognized Telegram user attempting to control the system triggers an instant alert to the authorized admin chat IDs.

Personalization

■ 120+ Background Themes

A huge built-in gallery of animated gradient themes to match any mood or room.

■ Custom Switch Icons

Pick from Light, Fan, AC, TV, Socket, Gate/Door, Water Pump, Heater, Speaker, Computer — or the default look.

■ Custom Switch Names

Rename every channel to match what it actually controls, reflected everywhere — web, Telegram, and Alexa.

■ Sound & Haptic Feedback

Satisfying click/save sound effects plus phone vibration on toggle — both independently mutable.

Engineering & Reliability

Built to run smoothly on the ESP8266's limited memory, 24/7.

■ Heavily RAM/Heap Optimized

Extensive use of fixed buffers over heap Strings, a trimmed TLS buffer, and reused static buffers minimize memory fragmentation.

■ Non-Blocking Debounce

Physical/touch switch inputs use a millis()-based debounce filter — the device never freezes waiting on a switch.

■ Persistent EEPROM Storage

WiFi, names, timers, Telegram/Alexa settings, and preferences all survive power loss.

■ Hardware Factory Reset

Hold the reset button for 5 seconds to wipe all settings and return to factory defaults — no computer required.

SUPERNOVA SMARTHOME

One firmware. Every way to control your home.
See the User Manual for full setup & wiring instructions.